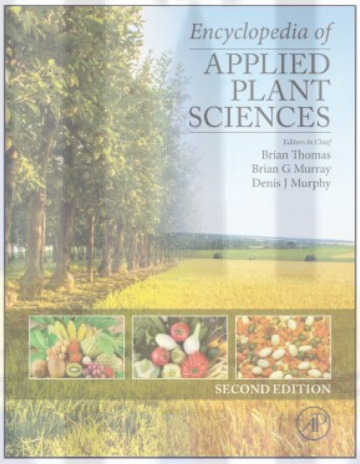

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Nematodes

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Nematodes

Nematodes or round worms are the most abundant multicellular animals on Earth. Free-living nematodes can be as long as 5 cm while animal parasitic nematodes may reach up to a length of 2 m. It is estimated that 1 million species exist in the phylum Nematoda of which 25 000 have been described so far. Ecologically, nematodes are ubiquitous, with species discovered in extreme environments such as gold mines in South Africa to polar regions. The life span of nematodes ranges from a few days for *Caenorhabditis elegans* to 15 years for the human parasitic hookworms *Necatoriasis americanus*. Based on phylogenetic analysis using the small subunit RNA genes of nematodes, the phylum Nematoda is divided into five clades with each including parasitic species (Figure 1).

While free-living nematodes help decompose dead matter and make nutrients available in soil for plants, parasitic nematodes cause economic and health problems worldwide. Animal parasitic nematodes include disease-causing worms of humans and animals parasitizing both invertebrates and vertebrates, including domestic animals and fish, reducing their health and leading to economic losses. Plant parasitic nematodes cost world agriculture an estimated $125 billion per annum.

Nematode Biology

Nematodes are structurally cylindrical, bilaterally symmetrical, and have digestive, nervous, excretory, and reproductive systems, but lack a circulatory system. The body is unsegmented and

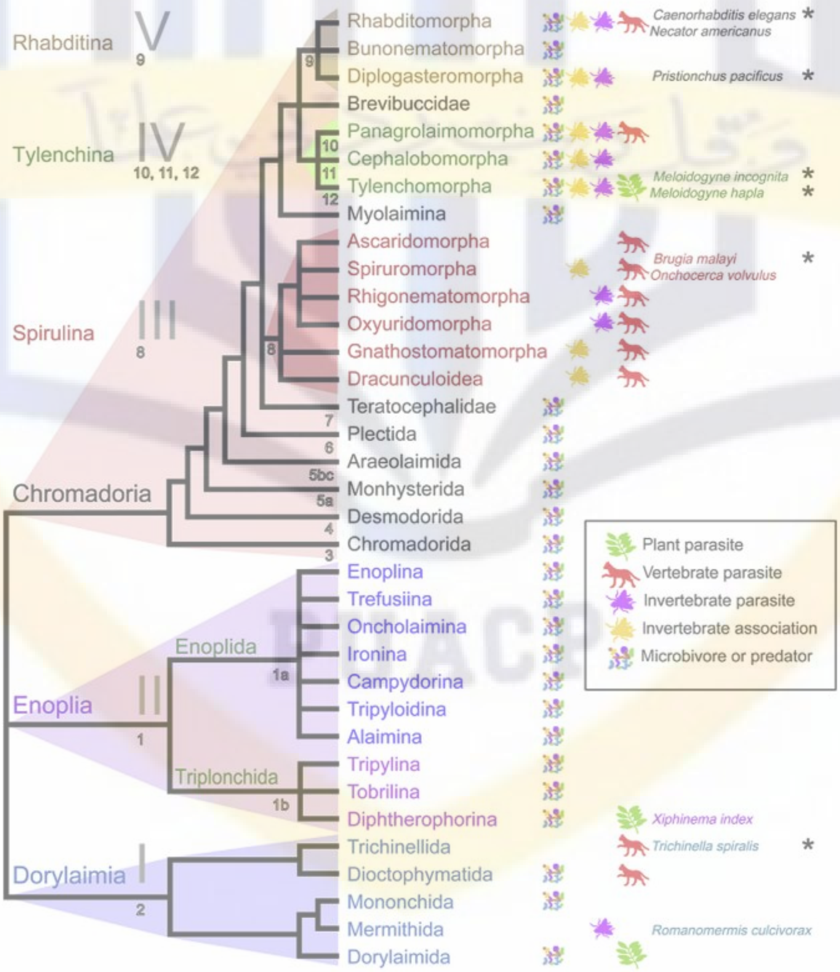

Figure 1 Phylogenetic relationship between nematode clades based on the small subunit ribosomal RNA gene. Parasitic nematodes are present in all clades, and their hosts are indicated pictorially.

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enveloped in a thick cuticle which protects the nematode from the external environment and forms a flexible exoskeleton. Beneath the epidermis is a layer of muscle cells that extend throughout the entire body enabling nematode movement. The oral cavity is typically a cuticle-lined capsule which may have modifications such as bristles, plates, or toothlike structures. Parasitic nematodes, and especially plant parasitic ones, have a sharp stylet which the animal thrusts into its prey. Digestive glands found in the pharynx produce enzymes that digest food. Entry into host plant tissues is achieved using the hollow mouth stylet with the aid of secretion of various proteins. In the case of plant hosts these may include cell wall–degrading enzymes, and for vertebrate parasitic nematodes proteases and hyaluronidases can help the nematodes penetrate and move while feeding from host cells.

At the anterior end of the body, the nerves branch from a dense circular nerve ring surrounding the pharynx which serves as a brain. Variations in morphological characteristics such as body shape, presence or absence of stylets, location of the vulva, tail shapes, spicules, cuticular ornamentation, etc. have been commonly used to identify nematodes morphologically. Most nematode species have separate male and female individuals, and reproduction is usually sexual. The eggs hatch into larvae which appear identical to the adults, but with an underdeveloped reproductive system. These juveniles go through several molts or ecdysis to become adults. Some nematodes are hermaphrodites and retain self-fertilized eggs inside the uterus until they hatch. The genus *Meloidogyne* of the plant parasitic group exhibits different modes of reproduction including sexual reproduction, facultative sexuality, meiotic and mitotic parthenogenesis. In some species males are rare (e.g., *Pratylenchus*).

Free-Living Nematodes

Free-living nematodes feed on bacteria, algae, fungi, dead organisms, and living tissues. They release nutrients for plant use and improve soil structure and water holding capacity. They are usually the most abundant type of nematodes in soil and marine environments. *Caenorhabditis elegans* is a free-living bacterial feeding nematode and is the most thoroughly studied nematode because it was chosen as a 'model' nematode species. *Caenorhabditis elegans* was selected by Sydney Brenner in 1965 as a model animal in particular to study animal development, genetics, and behavior. *Caenorhabditis elegans* was also the first multicellular organism to have its genome fully sequenced and annotated (see Relevant Website), and its genome contains about 20 000 genes. Terrestrial nematodes live and move in water films between soil particles and are sometimes used as indicators of soil health since they cannot survive in anaerobic or extreme soil conditions. It has been estimated that more than half of the nematode species existing are free-living forms. Marine nematodes are found at the ocean floor and continental shelves while some are restricted to thermal waters, for example, *Greenia* spp., *Dorylaimus* spp., and *Microlaimus* spp. Marine species feed mainly on organic matter in sand and gravel; some forms feed on algae or even other nematodes, which can contribute to controlling pest species.

Entomopathogenic Nematodes

Entomopathogenic nematodes are insect parasitic nematodes and include species from the genera *Heterorhabditis*, *Steinernema*, and *Beddingia*. These nematodes enter insect bodies through natural openings, and then kill their insect hosts and feed and reproduce on host tissue. They can be economically beneficial, and some species are available commercially as biological control for pests which damage plants and animals, and even against house termites. The nematodes develop and reproduce inside the host, and under optimum conditions nematodes exit the host body 7–15 days after infection and search for new hosts. Some species harbor mutualistic bacteria (e.g., *Xenorhabdus*, *Photorhabdus*), which are involved in killing the host insect and also feed from liquefying host tissues. The mutualistic bacteria also produce a range of antibiotics, which may have evolved to prevent competition for the host by other infective juveniles. The application of entomopathogenic nematodes as biocontrol agents for insect pests has been progressing rapidly because of their ease of mass production, broad host range, and safety to mammals and the environment.

Animal and Human Parasitic Nematodes

Animal parasitic nematodes infect animals from all habitats – there are about 342 species which infect humans. Humans worldwide are infected by nematodes each year, mostly in tropical areas and developing countries, in the order of 3.5 billion. Nematodes parasitic on humans include ascaris, filarial nematodes, hookworms, pinworms, and whipworms. Species include *Ancylostoma duodenale*, *Necator americanus*, *Trichinella spiralis*, *Wuchereria bancrofti*, *Onchocerca volvulus*, etc. They can attack the muscles, alimentary canal, eyes, and other body tissues. Entry can be through skin, ingestion of eggs in food, or through bites by an infected vector (e.g., mosquitoes that carry filarial worms). In some cases, animals may be intermediate hosts where the nematodes enter and grow for a period of time as larvae and then become dormant cysts. If a human eats the infected meat, the juveniles become active again and grow into adult worms. *Ascaris* and *Trichuris* spp. can survive dry conditions or unfavorable temperatures as eggs. Tapeworms also infect cows, fish, or pigs and then latch onto the intestinal wall of a human that consumes them. Heartworms causing heartworm disease infest hearts, arteries, and lungs of dogs and some cats.

Plant Parasitic Nematodes

Nematodes also include species that parasitize plants and reduce crop yield and quality. Plant parasitic nematode species are generally 250 μm to 12 mm in length and 15–35 μm in width. They can infest almost all parts of plants depending on the species: some can infest trees, such as the pine wilt nematode (*Bursaphelenchus* spp.); others can infest leaves (*Aphelenchoides* spp.) and stems of bulbs (*Ditylenchus* spp.). However, the most economically important impact on agriculture is caused by species that parasitize plant roots.

The lifestyles of plant parasitic nematodes (PPNs) can be ectoparasitic or endoparasitic as defined by their feeding habit (outside or within the root, respectively) and migratory or sedentary depending on the mobility after infection. These nematodes have also developed strategies and mechanisms to survive unfavorable conditions, for example, as eggs of cyst nematodes, which can survive in soil for many years. Migratory endoparasitic nematodes navigate their way from cell to cell damaging and ingesting cell contents and acquiring nutrients without inducing permanent feeding sites: most can move into and out of roots. Examples of migratory nematodes include the burrowing nematode (*Radopholus* spp.), root lesion nematodes (*Pratylenchus* spp.), pine wilt nematodes (*Bursaphelenchus* spp.), and the foliar nematode (*Aphelenchoides* spp.).

In contrast, sedentary endoparasitic nematodes invade plant roots and modify host cells to support feeding for 4–6 weeks. Examples of sedentary endoparasitic nematodes include root-knot nematodes (*Meloidogyne* spp.) and cyst nematodes (*Globodera* spp. and *Heterodera* spp.). Root-knot nematodes modify host gene expression and metabolism by inducing about six multinucleate giant cells from which they feed, and which are surrounded by galls or tissues giving the appearance of 'knots,' hence the name root-knot nematode. Cyst nematodes develop a multinucleate feeding cell called a syncytium, which forms by coalescence of a series of neighboring cells from which they feed. In both cases a 'feeding tube' develops in the cytoplasm when the nematodes feed; these act as pressure regulators and ultrafilters. The feeding structures induced by sedentary nematodes remain metabolically active until the nematodes complete their life cycles. The eggs of plant parasitic nematodes can remain dormant in soil for years and some hatch by sensing stimuli from a prospective host.

Control of Parasitic Nematodes

Chemical Control

For animal parasitic nematodes, several antihelminthic drugs have been developed and are used routinely to prevent nematode infections. Resistance as a result of variably susceptible populations has been reported against several antihelminthic drugs, for example, to ivermectin and moxidectin against *Haemonchus contortus* and *Ostertagia* spp.

In tropical and semitropical environments it is clear that some form of nematode management should be undertaken to combat plant parasitic nematodes, particularly in vegetable crop production, because of the pressures of nematode infestation. Methods of control include crop rotation, 'clean' cultivation, and chemical nematicides, although many of the latter, including aldicarb, oxamyl and methyl bromide, are regarded as environmentally hazardous, and restrictions in their use imposed by many governments resulted in limited or ended their use.

Other approaches to plant nematode control include applying antagonistic compounds that can block chemosensation employed by nematodes to locate host roots, but such effects are short-lived. Nitrogenous salts and ammonia released from decaying organic matter can also repel nematodes such as *Meloidogyne incognita*.

Biological Control

In an attempt to minimize environmental risks, biological methods have also been evaluated for control of plant parasitic nematodes. Nematodes are exposed to various predators in the soil, including fungi, bacteria, viruses, protozoans, insects, mites, and predatory nematodes. Obligate parasites of nematodes such as *Bacillus penetrans* and *Pasteuria penetrans* have been well studied as biological control agents: for root-knot nematodes the presence of *P. penetrans* in soil affects females developing in roots and reduces the production of egg masses. Although *Pasteuria* species have a narrow host range, species which specifically parasitize nematodes have been identified. For example, *Pasteuria thornei* can infect *Pratylenchus* spp. while *Pasteuria nishizawae* infects *Globodera* and *Heterodera* species. Many species of the genus *Bacillus* have also been tested for their efficacy as agents for nematode control, and *Bacillus firmus* is available in commercial formulations such as BioNem (3% lyophilized spores with 97% nontoxic additives), application of which results in a decrease in plant parasitic nematode populations and root infestation.

Various fungi parasitic to nematodes can also be used as biocontrol agents. These affect nematodes by trapping them, thus restricting their movement (e.g., *Arthrobotrys*, *Dactylellina*, *Drechslerella*), their ability to become endoparasites (e.g., *Haptocillium*, *Harposporium*, *Drechmeria*), feeding on eggs and females (e.g., *Pochonia*, *Paecilomyces*, *Lecanicillium*), and by toxin production (e.g., *Pleurotus*, *Coprinus*). Arbuscular mycorrhizal fungi have also been studied as potential biocontrol agents because their exudates can paralyze J2s of *M. incognita* and reduce their penetration into tomato roots. The nonpathogenic endophyte *Fusarium oxysporum* when inoculated on tomato plants induced resistance to *M. incognita* and resulted in a 45% reduction in nematode penetration. Predatory nematodes can also substantially reduce plant parasitic nematode populations. Predatory nematodes include mononchids, and dorylaimids are favored by improved soil texture. Suppression of plant parasitic nematode populations has been associated with higher soil organic matter and a greater spectrum of free-living nematode communities.

The effectiveness of biological control depends on many factors, including host plant genotype and parasitic nematode population. Plant rhizosphere exudates also affect microbial populations in soil, and for high infestation of parasitic nematodes, higher populations of biocontrol agents are required. Some recently developed biological nematicides are available commercially, but large-scale application in conditions which favor nematodes but not the biological agent can reduce their effectiveness – on a smaller horticultural scale some can be effective.

Phytochemicals have also been evaluated for control of plant nematodes, for example, polythienyls from marigold species may reduce populations of root-knot and root-lesion nematodes. Species of the family Asteraceae when used as soil amendments or as essential oil formulations can reduce reproduction of *Meloidogyne artiellia* by up to 96%. Lauric acid from the crown daisy repels *M. incognita* J2s and disrupts *Miflp18* expression, ultimately reducing parasitism. Intercropping of such plants has been used to control some nematodes, but most natural compounds are too expensive to synthesize, unstable or otherwise not economical or effective to use as control agents on a large scale.

Natural Resistance in Plants against Parasitic Nematodes

One aspect that confuses assessment of plant nematode damage is that the above-ground symptoms of infection are often similar to those of nutrient deficiency (stunting, chlorosis, and wilting) and the extent of damage is often underestimated. The most practical method to control plant nematodes is the use of resistant plants. Natural resistance genes often derived from related wild species, which have been introduced through conventional breeding into crops to protect them against nematodes, have been used commercially. Examples include, potato cultivars with the *H1* resistance gene derived from *Solanum andigena* which has been durable and effective against *Globodera rostochiensis*.

However, most resistance genes are not broad spectrum and are not available for all commercial crops. For example, the *Mi* gene of tomato confers resistance against *M. incognita* and *Meloidogyne hapla*, but is not effective against *Meloidogyne javanica* – virulent strains of nematodes can infect these resistant cultivars having *Mi* gene. To overcome this difficulty, a new strategy, based on the RNA interference (RNAi or 'gene silencing') is being investigated, which may be used to target conserved genes and could potentially result in broad spectrum environmentally safe nematode control.

RNA Interference

Since its discovery in *C. elegans*, RNAi has become the only widely amenable reverse genetics tool for parasitic nematodes. RNAi is an endogenous gene regulation mechanisms in eukaryotes and can be harnessed to target messenger RNA by introducing a complementary trigger, i.e., double-stranded RNA (dsRNA) or short interfering RNA (siRNA) hence preventing or downregulating expression of a particular protein. This technique has been used to study gene functions in *C. elegans* on a large scale but has proved to be challenging in parasitic nematodes. The main reasons being obligate parasitism, small size which makes microinjection inapplicable, and lack of parasitic nematode transformation methods. For animal parasitic nematodes, electroporation was tested as a possible means of delivery mechanism for dsRNA, but high levels of larval death meant that this method was inappropriate.

However, a soaking method where nematodes are incubated in a solution containing dsRNA or siRNA has been developed where nematodes are stimulated to take up external solution containing dsRNA by the addition of the neurostimulants such as octopamine, serotonin, and resorcinol which induce pharyngeal pumping or solution ingestion.

While RNAi is being successfully applied to study gene functions in different organisms, in parasitic nematodes, it has also been exploited as a control measure by silencing genes essential for survival, development, or parasitism. In animal parasitic nematodes, the effectiveness of RNAi appears to depend on the stability and level of transcript as well as the site of gene expression. Genes expressed in intestine, excretory cells, and amphids displayed a higher success rate of RNAi. Silencing of target genes by soaking of plant parasitic nematodes in dsRNA reduces viability, reproduction, and ultimately parasitism. This has led to the development of host-induced RNAi, in which plants are genetically modified to produce dsRNA against nematode gene targets, so far with promising results with a range of different targets. Table 1 indicates studies on nematodes undertaken using host-induced gene silencing.

Table 1 A summary of experimental studies using host-induced gene silencing to control PPNs

Plant	*Gene targeted*	*Nematode species*	*Effect of host delivered knockdown*
Arabidopsis	Secreted peptide (*16D10*)	*M. incognita*	39–83% reduction in the number of eggs per gram of root, 63–90% reduction in the number of galls, general decrease in gall size
	Secreted peptide (*16D10*)	*M. javanica*	39–83% reduction in the number of eggs per gram of root, 63–90% reduction in the number of galls, general decrease in gall size
	Secreted peptide (*16D10*)	*M. arenaria*	39–83% reduction in the number of eggs per gram of root, 63–90% reduction in the number of galls, general decrease in gall size
	Secreted peptide (*16D10*)	*M. hapla*	39–83% reduction in the number of eggs per gram of root, 63–90% reduction in the number of galls, general decrease in gall size
Arabidopsis	Secreted peptide (*16D10L*)	*M. chitwoodi*	68–74% reduction in egg mass production
	Nematode effector (*NULG1a*)	*M. javanica*	88% reduction in infection
	Ubiquitin-like protein (*4G06*)	*H. schachtii*	23–64% reduction in developed females
	Cellulose binding protein (*3B05*)		12–47% reduced infection
	SKP1-like protein (*8H07*)		>50% reduced infection
	Zinc finger protein (*10A06*)		42% reduced infection
	Nematode secreted peptide (*Hssyv46*)		36% reduced cyst formation
	Nematode secreted peptide (*Hs5d08*)		20% reduced cyst formation
	Nematode secreted peptide (*Hs4e02*)		20% reduced cyst formation
	Nematode secreted peptide (*Hs4F01*)		55% reduced cyst formation
	Parasitism effector (*30C02*)		92% reduced cyst formation
	Parasitism gene (*Mi8D05*)	*M. incognita*	Up to 90% reduction in infection

Table 1 A summary of experimental studies using host-induced gene silencing to control PPNs—cont'd

Plant	*Gene targeted*	*Nematode species*	*Effect of host delivered knockdown*
Tobacco	SNF chromatin remodeling complex component (*snfc-5*)	*M. incognita*	>90% reduction in infection
	Pre-mRNA splicing factor (*prp-21*)		>90% reduced infection
	Putative transcription factor (*MjTis11*)	*M. javanica*	Decreased transcript level in feeding nematodes
	FMRFamide-like gene (*flp-14*)	*M. incognita*	Upto 86% reduction in reproduction rate
	FMRFamide-like gene (*flp-18*)		Upto 82% reduction in reproduction rate
	Serine protease gene (*ser-1*)		30% reduced egg production
	Cysteine protease (*cpl-1*)		42% reduced egg production
Soybean	Major sperm protein	*H. glycines*	68% reduction in cyst formation
	Ribosomal protein 3a (*rps-3a*)		87% reduced cyst formation
	Ribosomal protein 4 (*rps-4*)		81% reduced cyst formation
	Spliceosomal SR protein (*spk-1*)		88% reduced cyst formation
	Synaptobrevin (*snb-1*)		93% reduced cyst formation
	Beta subunit of the COPI complex (*Y25*)		81% reduced cyst formation
Soybean	Pre-mRNA splicing factor (*prp-17*)	*H. glycines*	79% reduction in infection
	Uncharacterized protein (*cpn-1*)		95% reduction in infection
	L-Lactate dehydrogenase	*M. incognita*	57% reduced infection, 77% reduction in nematode diameter
	Mitochondrial stress-70 protein		2% reduction in infection, 8% reduced nematode diameter
	ATP synthase beta-chain mitochondrial precursor		64% reduced infection, 62% reduced nematode diameter
	Tyrosine phosphatase		95% reduced infection, 82% reduced nematode diameter
Tomato	Troponin C (*snc*)	*M. incognita*	59% reduced hatching from eggs
	Calreticulin (*crt*)		No effect after silencing. Reduced infection by the progeny
	Dual oxidase (*duox*)		61% reduced infection, 52% reduction in saccate nematodes
	Signal peptidase complex 3 (*spc3*)		52% reduction in infection, 63% reduction in saccate nematodes
	Subunit of 19S regulatory complex (*rpn-7*)		Up to 66.5% reduction in infection
	Subunit of 19S regulatory complex (*rpn-7*)		50.8% reduction in egg production
Potato	Secreted peptide (*16D10L*)	*M. chitwoodi*	44–56% reduction in egg mass production
Grape (hairy root)	(*16D10*)	*M. incognita*	Significant reduction in egg production

With more genomes of parasitic nematodes being sequenced, transgenic nematode control strategies are likely to become more sophisticated and specific in delivering an economically feasible control strategy against these pests.

See also: **Crop Diseases and Pests:** Integrated Pest Management: Practice. **Plants and the Environment:** The Rhizopshere and Its Microorganisms.

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Relevant Website

www.wormbase.org – Explore Worm Biology (last accessed 29.05.16).